Astrocyte expression in Alzheimer's disease: a donor-level reanalysis

Zilong Wang | Université Paris-Saclay, Master 2 Bioinformatics & Biostatistics

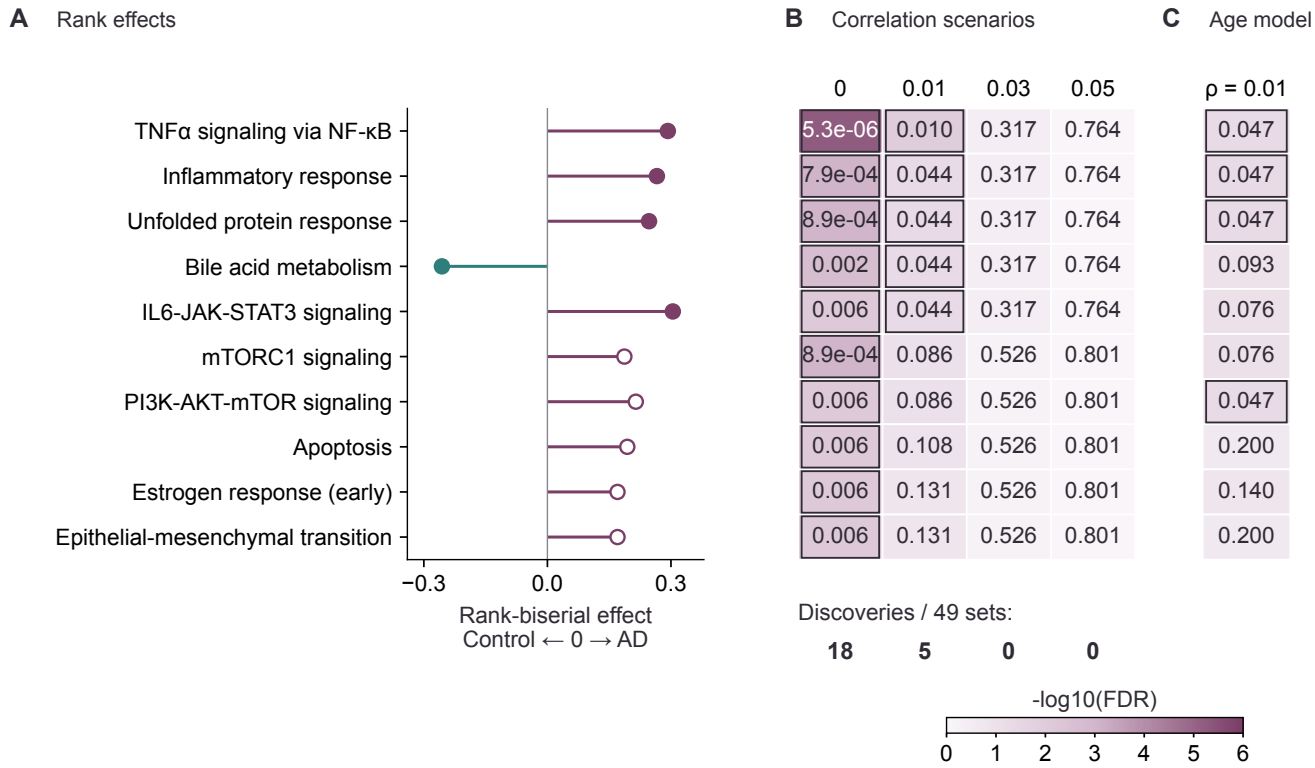
Research project summary | Project period: August 2026

Research question and approach

Using public GSE174367 data, I combined counts from 4,756 astrocyte nuclei into profiles for 11 AD and 7 control donors. I compared the groups with edgeR, then checked how age adjustment, gene correlation assumptions and donor omissions affected the results. The donor, rather than each nucleus, was the biological replicate.

Findings

No individual gene passed correction among 12,331 tested genes. Inflammatory and protein-stress gene sets ranked toward AD. At the default correlation of 0.01, 5 of 49 Hallmark sets met BH FDR < 0.05. Age adjustment retained 3 of the original five; none remained significant at correlations 0.03 or 0.05. All five kept their direction across 18 donor omissions, but each met FDR < 0.05 in only 9-11 omissions.



Top 10 primary-P-ranked sets. Filled points / outlines: FDR < 0.05.
B: fixed ranks. C: recorded-age model with a refitted ranking.

Figure | Program ranking and statistical support. A, Primary rank-biserial effects; filled points meet default FDR < 0.05. B, FDR across four assumed correlations, holding ranks fixed. C, The recorded-age model. Outlined cells meet the same FDR threshold; color shows evidence, not pathway activity.

Interpretation. The gene-set rankings describe this cohort, with statistical support that depends on the model. Age and sex imbalance, capped ages and conflicting batch labels limit the comparison; donor-level expression also mixes cell-state proportions with expression within states. The overlapping donor omissions do not provide independent validation.

Data and code. Morabito et al., Nature Genetics (2021), doi:10.1038/s41588-021-00894-z; GEO GSE174367. Code, complete tables and supplementary methods accompany the report.